

B.E / B.Tech. PRACTICAL END SEMESTER EXAMINATIONS, APRIL/MAY 2026
Fourth Semester

CS23421- Database Management Systems Laboratory
(Regulations 2023)

Time: 3 Hours		Answer any one Question			Max. Marks 100	
Aim & Algorithm	Queries/ Program	Execution	Output	Record	Viva	Total
10	40	20	10	10	10	100
1. Design and implement a College Examination System with student, subject, and marks tables. Normalize up to 3NF, enforce constraints, and generate result reports.						
2. Given: EMPLOYEE(EID, NAME, DEPT, SALARY) PROJECT(EID, PID, HOURS) Write SQL queries to find employees working on multiple projects, highest salary per department, and average hours worked.						
3. Develop a Clinic Management System. Implement patient records, doctor assignments, triggers for appointment logging, and generate reports.						
4. Given: CUSTOMER(CID, NAME, CITY) ORDERS(OID, CID, AMOUNT) Write queries to find top customers, customers with no orders, and city-wise revenue.						
5. Design a Digital Wallet System. Implement transactions with TCL commands and ensure ACID properties.						
6. Given SALES(SID, PRODUCT, AMOUNT, REGION): Perform region-wise aggregation, identify best-selling products, and calculate average revenue.						
7. Design a School Management System with attendance tracking and grading logic using PL/SQL.						
8. Given PRODUCT(PID, NAME, PRICE): Write queries using subqueries to find products above average price and least expensive products per category.						
9. Design an E-Commerce Order System. Implement order workflows using stored procedures and triggers.						
10. Given STUDENT, CLASS, ENROLL tables: Write join queries to retrieve student-class mappings and identify unassigned students.						
11. Design a Staff Leave Management System. Implement validation using triggers and generate leave reports.						

12. Given TRANSACTION(TID, AID, AMOUNT):

Create triggers for auditing and prevent invalid transactions.

13. Design a Movie Ticket Booking System. Implement constraints, triggers, and analyze booking trends.

14. Given the following ORDER table with a large number of records:

ORDERS(OID, CID, PRODUCT_ID, ORDER_DATE, AMOUNT, STATUS)

Perform the following:

Create appropriate indexes to improve query performance

Evaluate query performance before and after indexing

Justify the optimization techniques used

15. Given the following tables:

CUSTOMER(CID, NAME, CITY)

PAYMENT(PID, CID, AMOUNT, PAYMENT_DATE)

Perform the following:

Write SQL queries to find customers with total payments above a threshold

Identify customers who made no payments

Create indexes to optimize payment-related queries

Analyze performance improvement

16. Given the table:

EMPLOYEE(EID, NAME, DEPT, SALARY, JOIN_DATE)

Perform the following:

Write queries to find department-wise average salary and recent joiners

Create appropriate indexes for salary and date-based queries

Compare performance before and after indexing

Justify your indexing strategy

17. Given the tables:

STUDENT(SID, NAME, DEPT)

ATTENDANCE(SID, DATE, STATUS)

Perform the following:

Write queries to calculate attendance percentage for each student

Identify students with attendance below a threshold

Create a view for attendance summary

Optimize queries using indexing

18. Given the table:

PRODUCT(PID, NAME, CATEGORY, PRICE, STOCK)

Perform the following:

Write queries to identify top-selling categories (assume sales logic if needed)

Find products with low stock levels

Create indexes on frequently queried columns

Analyze and justify performance improvements

19. Develop a Firebase-based chat application with real-time updates.

20. Perform analytical queries on sales dataset to identify trends and seasonal patterns.

Internal Examiner

External Examiner